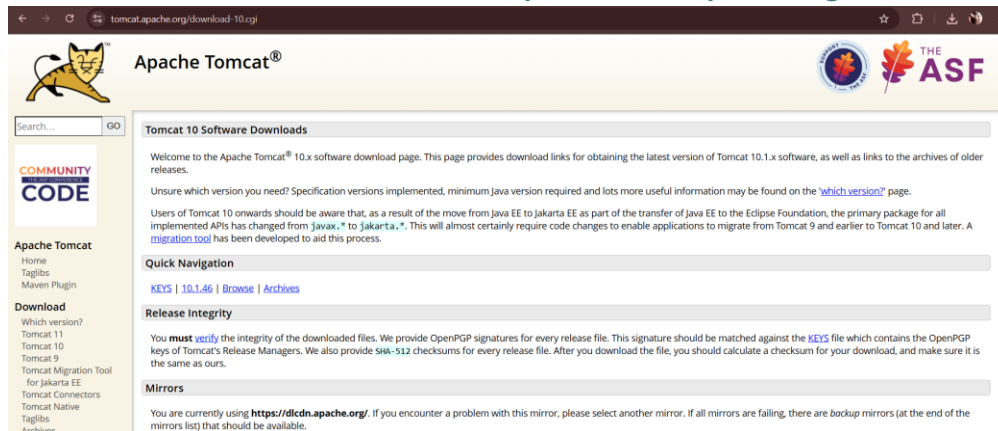


Aim: Learning JSP programs.

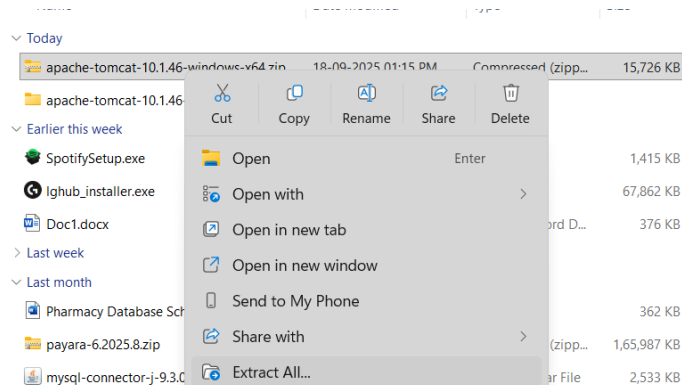
Steps for Running the JSP program:

Steps:

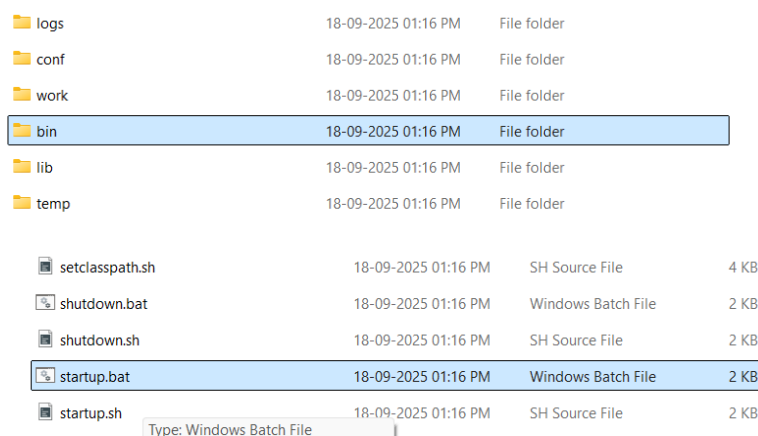
1.Download the Tomcat API here is the link: <https://tomcat.apache.org/download-10.cgi>



2.Unzip the Tomcat file.



3.Open the Tomcat API then go to bin then click startup.bat (make sure that the JAVA_HOME variable is set up in Environment variables so that Tomcat can easily access the JDK) this make the Tomcat alive.

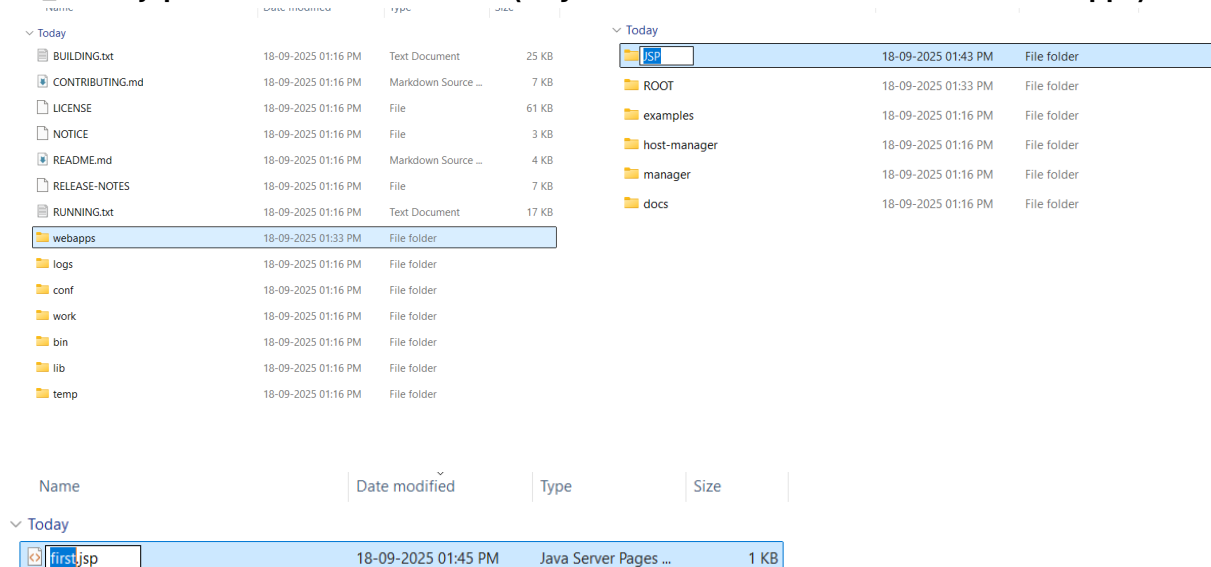


```

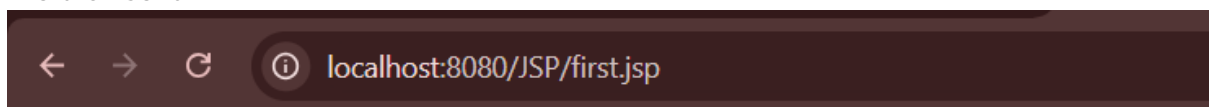
Tomcat
n directory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\docs] has finished
in [228] ms
18-Sep-2025 13:32:51.099 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application di
rectory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\examples]
18-Sep-2025 13:32:51.489 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web applicatio
n directory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\examples] has fini
shed in [390] ms
18-Sep-2025 13:32:51.489 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application di
rectory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\host-manager]
18-Sep-2025 13:32:51.519 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web applicatio
n directory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\host-manager] has fini
shed in [38] ms
18-Sep-2025 13:32:51.519 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application di
rectory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\manager]
18-Sep-2025 13:32:51.546 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web applicatio
n directory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\manager] has fini
shed in [27] ms
18-Sep-2025 13:32:51.547 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application di
rectory [C:\Users\ROHIT\Downloads\apache-tomcat-10.1.46-windows-x64\apache-tomcat-10.1.46\webapps\ROOT]

```

4. Now go to the webapps and then create a folder of your name. Then create a file `file_name.jsp` inside the current folder (i.e. your new created folder inside the webapps)



5. Execute your jsp file by writing http://localhost:8080/folder_name/file_name.jsp inside the browser url.



CODE:

```

<html>
<body>
    <% out.println("Welcome Mr.Atomic <br>"); %>
    <% out.println("If I don't want to be vaporized in a nuclear explosion, I simply have to
    become nuclear myself....I am Atomic."); %>
</body>
</html>

```

OUTPUT:

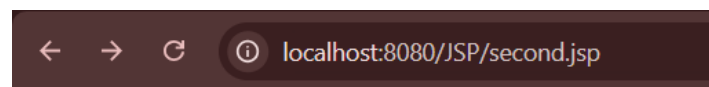
Welcome Mr.Atomic
If I don't want to be vaporized in a nuclear explosion, I simply have to become nuclear myself....I am Atomic.

1. Write a JSP program that accepts a User Name and Age from a HTML form and display the same.

CODE:

```
<html>
<body>
<%
    String name = request.getParameter("uname");
    String age = request.getParameter("age");
    if (name == null || age == null){
%>
        <h2>Enter Your Details</h2>
        <form method="post" action="second.jsp">
            User Name: <input type="text" name="uname"><br><br>
            Age: <input type="number" name="age"><br><br>
            <input type="submit" value="Submit">
        </form>
<%
    }
    else{
        out.println("Welcome " + name + "<br>");
        out.println("Your Age is: " + age);
    }
%>
</body>
</html>
```

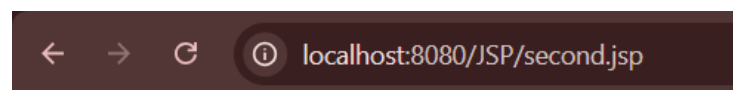
OUTPUT:



Enter Your Details

User Name:

Age:



Welcome Mr.Atomic
Your Age is: 19

2. Write a JSP code that accepts username and password from HTML file and validates the user.

CODE:

```
<html>
<body>
<%
    String uname = request.getParameter("uname");
    String pass = request.getParameter("pass");
    if (uname == null || pass == null){
%>
        <h2>Login Page</h2>
        <form method="post" action="third.jsp">
            Username: <input type="text" name="uname" required><br><br>
            Password: <input type="password" name="pass" required><br><br>
            <input type="submit" value="Login">
        </form>
<%
    }
    else{
        String validUser = "Mr.Atomic";
        String validPass = "8169";

        if (uname.equals(validUser) && pass.equals(validPass)) {
            out.println("<h2>Welcome, " + uname + "!</h2>");
        }
        else{
            out.println("<h2 style='color:red'>Invalid Username or Password!</h2>");
            out.println("<a href='third.jsp'>Try Again</a>");
        }
    }
%>
</body>
</html>
```

OUTPUT:



← → ↻ ⓘ localhost:8080/JSP/third.jsp

Login Page

Username:

Password:

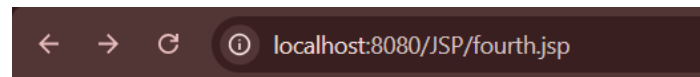
Welcome, Mr.Atomic!

Invalid Username or Password!

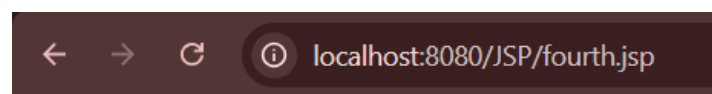
[Try Again](#)

3. Write a JSP code to print factorial of given number.**CODE:**

```
<html>
<body>
<%
    String numStr = request.getParameter("number");
    if (numStr == null){
%>
        <h2>Enter a Number</h2>
        <form method="post" action="fourth.jsp">
            Number: <input type="number" name="number" required>
            <input type="submit" value="Calculate">
        </form>
<%
    }
    else{
        int n = Integer.parseInt(numStr);
        long fact = 1;
        for (int i = 1; i <= n; i++){
            fact *= i;
        }
        out.println("<h2>Factorial of " + n + " is: " + fact + "</h2>");
    }
%>
</body>
</html>
```

OUTPUT:**Enter a Number**

Number:

**Factorial of 5 is: 120**